	Politechnika Bydgoska im. J. J. Śniadeckich Wydział Telekomunikacji, Informatyki i Elektrotechniki		
Przedmiot	Projektowanie serwisów sieciowych		
Prowadzący	mgr inż. Leonid Rusanov		
Temat	Wprowadzenie do Laravel		
Student (+ nr indeksu)	Dawid Wendt 122652		
Nr lab.	6	Data oddania spr.	3.12.2025

1. Cel ćwiczenia

Celem ćwiczenia jest zapoznanie się z frameworkiem Laravel, instalacją, strukturą projektu oraz tworzeniem podstawowych komponentów aplikacji, w tym klas modeli i widoków Blade.

2. Przebieg

```

Administrator: Wiersz polecenia
Microsoft Windows [Version 10.0.26100.7171]
(c) Microsoft Corporation. Wszelkie prawa zastrzeżone.

C:\Windows\System32>php --version
PHP 8.2.12 (cli) (built: Oct 24 2023 21:15:15) (ZTS Visual C++ 2019 x64)
Copyright (c) The PHP Group
Zend Engine v4.2.12, Copyright (c) Zend Technologies

C:\Windows\System32>composer --version
Composer version 2.9.2 2025-11-19 21:57:25
PHP version 8.2.12 (C:\xampp\php\php.exe)
Run the "diagnose" command to get more detailed diagnostics output.

C:\Windows\System32>node --version
v22.20.0

C:\Windows\System32>npm --version
10.9.3

C:\Windows\System32>

```

Zainstalowane wymagane rzeczy.

```
Wybierz Administrator: Wiersz polecenia

- Installing sebastian/recursion-context (4.0.3): Extracting archive
- Installing sebastian/object-reflector (4.0.1): Extracting archive
- Installing sebastian/object-enumerator (5.0.1): Extracting archive
- Installing sebastian/global-state (2.0.2): Extracting archive
- Installing sebastian/exporter (6.3.2): Extracting archive
- Installing sebastian/environment (7.2.1): Extracting archive
- Installing sebastian/dir (6.0.2): Extracting archive
- Installing sebastian/comparator (6.3.2): Extracting archive
- Installing sebastian/code-unit (3.0.2): Extracting archive
- Installing sebastian/call-parser (3.0.2): Extracting archive
- Installing phpunit/php-timer (7.0.1): Extracting archive
- Installing phpunit/php-test-template (4.0.1): Extracting archive
- Installing phpunit/php-invoker (5.0.1): Extracting archive
- Installing phpunit/php-file-iterator (5.1.0): Extracting archive
- Installing theseer/tokenizer (1.3.3): Extracting archive
- Installing sebastian/lines-of-code (3.0.1): Extracting archive
- Installing sebastian/complexity (4.0.1): Extracting archive
- Installing sebastian/code-unit-reverse-lookup (4.0.1): Extracting archive
- Installing phpunit/php-code-coverage (11.0.11): Extracting archive
- Installing phar-io/version (3.2.1): Extracting archive
- Installing phar-io/manifest (2.0.4): Extracting archive
- Installing myclabs/deep-copy (1.13.4): Extracting archive
- Installing phpunit/phpunit (11.5.44): Extracting archive

50 package suggestions were added by new dependencies, use 'composer suggest' to see details.
Generating autoloaders
> illuminateFoundationComposerScripts::postAutoloadDump
> @php artisan package:discover --ansi

INFO Discovering packages.

laravel/pail ..... DONE
laravel/sail ..... DONE
laravel/tinker ..... DONE
nesbot/carbon ..... DONE
nunomaduro/collision ..... DONE
nunomaduro/termwind ..... DONE

51 packages you are using are looking for funding.
See the composer fund command to find out more!
> @php artisan vendor:publish --tag=laravel-assets --ansi --force

INFO No publishable resources for tag [laravel-assets].

No security advisories found!
> @php artisan key:generate --ansi

INFO Application key set successfully.

> @php -r "file_exists('database/database.sqlite') || touch('database/database.sqlite');"
> @php artisan migrate --graceful --ansi

INFO Preparing database.

Creating migration table ..... 6.77ms DONE

INFO Running migrations.

0001_01_01_000000_create_users_table ..... 16.46ms DONE
0001_01_01_000001_create_cache_table ..... 4.82ms DONE
0001_01_01_000002_create_jobs_table ..... 13.13ms DONE

C:\Users\David\PycharmProjects\IPSS_David_Wendt\Zad_0>
```

Gotowy stworzony projekt

```

1 APP_NAME=Laravel
2 APP_ENV=local
3 APP_KEY=base64:Kam9wFtRUnhCmAB+0u4TaTeQ3ySv1eqfram04yP7a7I=
4 APP_DEBUG=true
5 APP_URL=http://localhost
6
7 APP_LOCALE=en
8 APP_FALLBACK_LOCALE=en
9 APP_FAKER_LOCALE=en_US
10
11 APP_MAINTENANCE_DRIVER=file
12 # APP_MAINTENANCE_STORE=database
13
14 # PHP_CLI_SERVER_WORKERS=4
15
16 BCRYPT_ROUNDS=12
17
18 LOG_CHANNEL=stack
19 LOG_STACK=single
20 LOG_DEPRECATIONS_CHANNEL=null
21 LOG_LEVEL=debug
22
23 DB_CONNECTION=sqlite
24 # DB_HOST=127.0.0.1
25 # DB_PORT=3306
26 # DB_DATABASE=laravel
27 # DB_USERNAME=root
28 # DB_PASSWORD=
29
30 SESSION_DRIVER=database
31 SESSION_LIFETIME=120
32 SESSION_ENCRYPT=false
33 SESSION_PATH=/
34 SESSION_DOMAIN=null
35
36 BROADCAST_CONNECTION=log
37 FILESYSTEM_DISK=local
38 QUEUE_CONNECTION=database
39
40 CACHE_STORE=database
41 # CACHE_PREFIX=
42
43 MEMCACHED_HOST=127.0.0.1
44
45 REDIS_CLIENT=phpredis
46 # REDIS_HOST=127.0.0.1
47 # REDIS_PASSWORD=null
48 # REDIS_TTL=0
49
50
```

Nowo wygenerowany klucz

```

Wybierz Administrator: Wierzy polecenia
- Installing thesauri/composer (4.3.1): Extracting archive
- Installing sebastian/lines-of-code (3.0.1): Extracting archive
- Installing sebastian/complexity (4.0.1): Extracting archive
- Installing sebastian/code-unit-reverse-lookup (4.0.1): Extracting archive
- Installing phpunit/php-code-coverage (11.0.11): Extracting archive
- Installing phar-io/version (3.2.1): Extracting archive
- Installing phar-io/manifest (4.0.4): Extracting archive
- Installing myclabs/deep-copy (1.13.4): Extracting archive
- Installing phpunit/phpunit (11.5.4): Extracting archive
59 package suggestions were added by new dependencies, use "composer suggest" to see details.
Generating optimized autoload files
 illuminate/foundation/composerScripts::postAutoloadDump
> @php artisan package:discover --ansi

[INFO] Discovering packages.

laravel/pail ..... DONE
laravel/sail ..... DONE
laravel/tinker ..... DONE
nesbot/carbon ..... DONE
nunomaduro/collision ..... DONE
nunomaduro/termwind ..... DONE

81 packages you are using are looking for funding.
see the "composer fund" command to find out more
> @php artisan vendor:publish --tag=laravel-assets --ansi --force

[INFO] No publishable resources for tag [laravel-assets].

No security vulnerability advisories found.
> @php artisan key:generate --ansi

[INFO] Application key set successfully.

> @php -r "file_exists('database/database.sqlite') || touch('database/database.sqlite');"
> @php artisan migrate --graceful --ansi

[INFO] Preparing database.

Creating migration table ..... 7.65ms DONE

[INFO] Running migrations.

0001_01_01_000000_create_users_table ..... 17.41ms DONE
0001_01_01_000001_create_cache_table ..... 5.11ms DONE
0001_01_01_000002_create_jobs_table ..... 12.51ms DONE

C:\Users\David\PycharmProjects\PS5_David_Wendt\Iad_6xcd lab06-app
C:\Users\David\PycharmProjects\PS5_David_Wendt\Iad_6\lab06-app>php artisan optimize:clear

[INFO] Clearing cached bootstrap files.

config ..... 2.14ms DONE
cache ..... 20.84ms DONE
compiled ..... 2.82ms DONE
events ..... 1.83ms DONE
routes ..... 0.93ms DONE
views ..... 10.58ms DONE

C:\Users\David\PycharmProjects\PS5_David_Wendt\Iad_6\lab06-app

```

```

Wybierz Administrator: Wierzy polecenia

> @php -r "file_exists('database/database.sqlite') || touch('database/database.sqlite');"
> @php artisan migrate --graceful --ansi

[INFO] Preparing database.

Creating migration table ..... 7.65ms DONE

[INFO] Running migrations.

0001_01_01_000000_create_users_table ..... 17.41ms DONE
0001_01_01_000001_create_cache_table ..... 5.11ms DONE
0001_01_01_000002_create_jobs_table ..... 12.51ms DONE

C:\Users\David\PycharmProjects\PS5_David_Wendt\Iad_6xcd lab06-app
C:\Users\David\PycharmProjects\PS5_David_Wendt\Iad_6\lab06-app>php artisan optimize:clear

[INFO] Clearing cached bootstrap files.

config ..... 2.14ms DONE
cache ..... 20.84ms DONE
compiled ..... 2.82ms DONE
events ..... 1.83ms DONE
routes ..... 0.93ms DONE
views ..... 10.58ms DONE

C:\Users\David\PycharmProjects\PS5_David_Wendt\Iad_6\lab06-app>php artisan migrate:fresh

Dropping all tables ..... 7.91ms DONE

[INFO] Preparing database.

Creating migration table ..... 6.85ms DONE

[INFO] Running migrations.

0001_01_01_000000_create_users_table ..... 17.19ms DONE
0001_01_01_000001_create_cache_table ..... 5.04ms DONE
0001_01_01_000002_create_jobs_table ..... 12.60ms DONE

C:\Users\David\PycharmProjects\PS5_David_Wendt\Iad_6\lab06-app>php artisan migrate:fresh --seed

Dropping all tables ..... 6.92ms DONE

[INFO] Preparing database.

Creating migration table ..... 6.87ms DONE

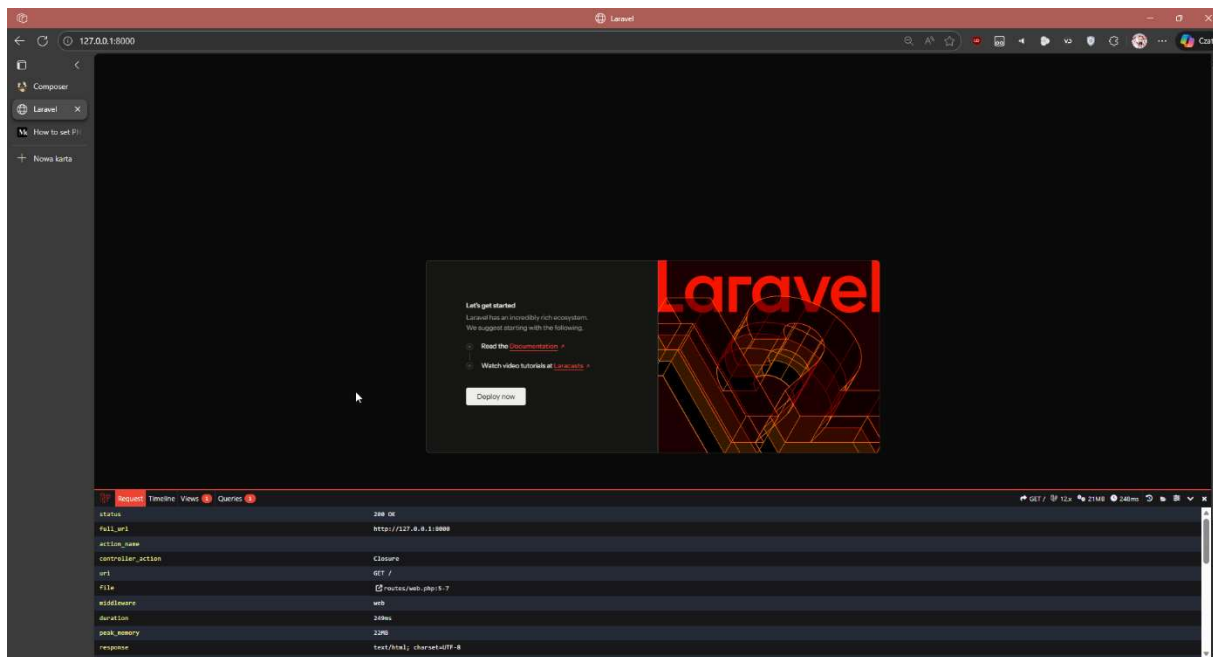
[INFO] Running migrations.

0001_01_01_000000_create_users_table ..... 15.51ms DONE
0001_01_01_000001_create_cache_table ..... 5.26ms DONE
0001_01_01_000002_create_jobs_table ..... 13.90ms DONE

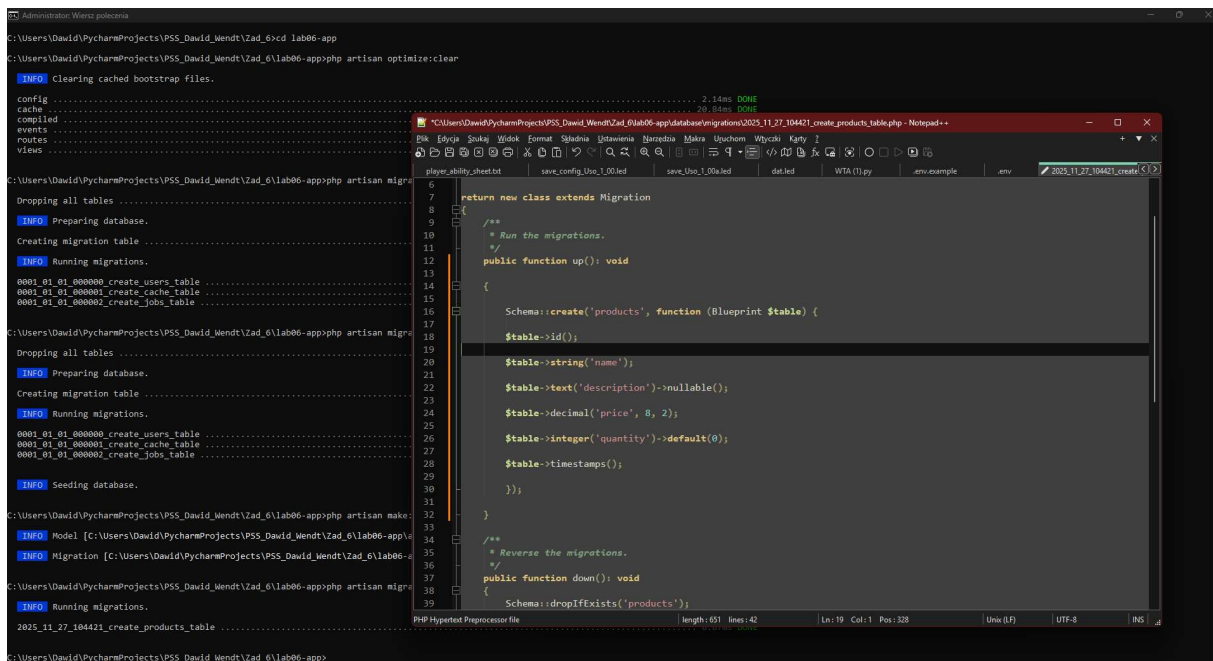
[INFO] Seeding database.

C:\Users\David\PycharmProjects\PS5_David_Wendt\Iad_6\lab06-app

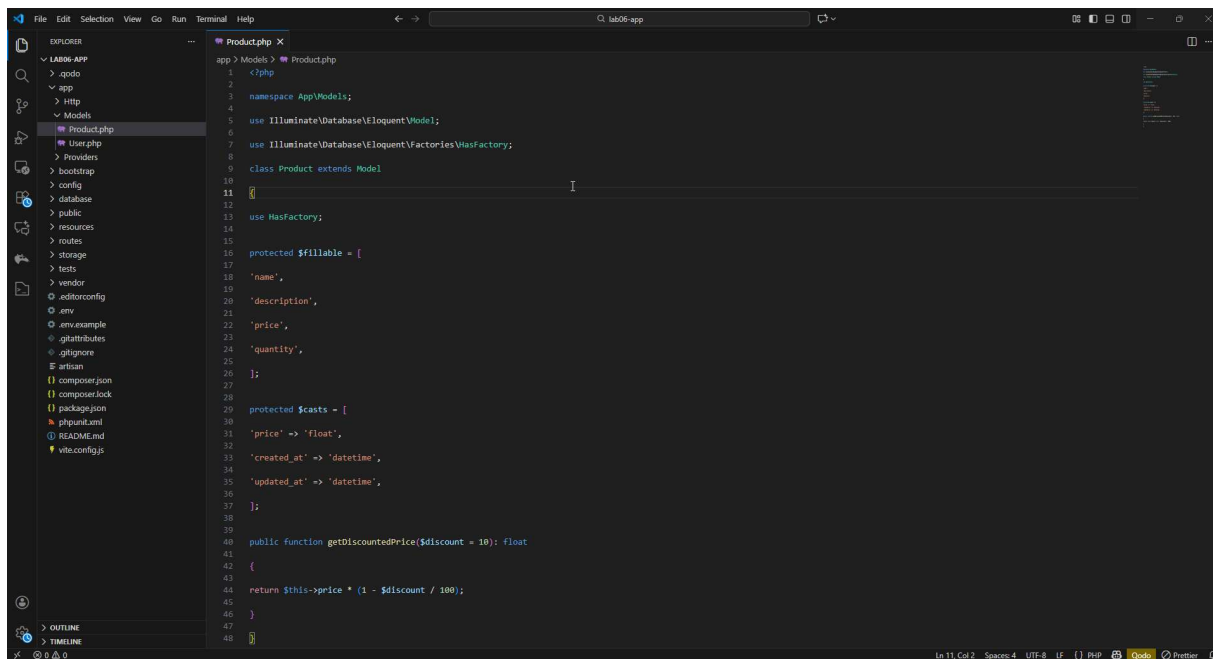
```



Widok po uruchomieniu Laravel wraz z ustawionym DEBUGBAR_ENABLED=true



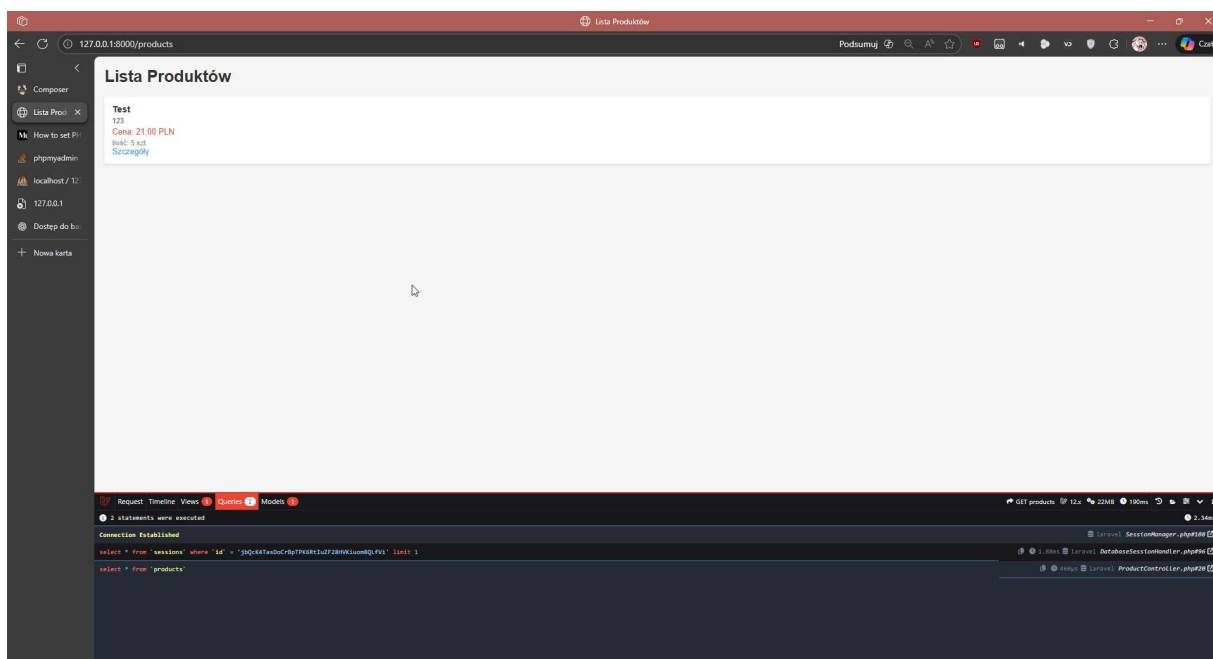
Kolejno edytowana funkcja up() po czym uruchomiona została migracja

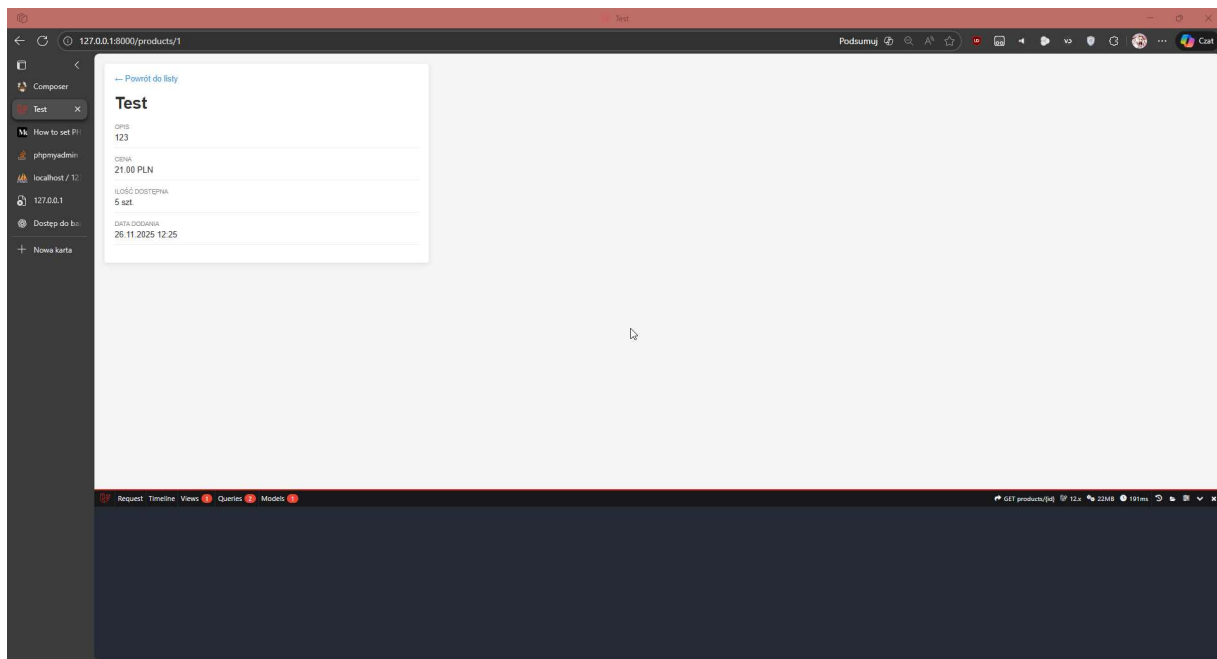


The screenshot shows a code editor with the file explorer on the left displaying a project structure. The main editor window shows the `Product.php` file. The code defines a `Product` class extending `Model`, with attributes `name`, `description`, `price`, and `quantity`. It also includes a `getDiscountedPrice` method.

```
1 app > Models > Product.php
2
3 <?php
4
5 namespace App\Models;
6
7 use Illuminate\Database\Eloquent\Model;
8
9 class Product extends Model
10 {
11     //
12
13     use HasFactory;
14
15     protected $fillable = [
16         'name',
17         'description',
18         'price',
19         'quantity',
20     ];
21
22     protected $casts = [
23         'price' => 'float',
24         'created_at' => 'datetime',
25         'updated_at' => 'datetime',
26     ];
27
28     public function getDiscountedPrice($discount = 10): float
29     {
30         return $this->price * (1 - $discount / 100);
31     }
32 }
```

Uzupełnione atrybuty oraz widoki po czym osiągniemy widok poniżej po dodaniu przykładowego produktu w np. phpmyadmin w zależności na jaką bazę danych się zdecydowaliśmy





Jak widać wszystko działa poprawnie po kliknięciu w produkt.

3. Wnioski

Na przyszłość należy zabezpieczać widoki przed pustymi danymi w bazie.

Przy niewielkiej edycji kodu można było otrzymać gotowe api gdzie tylko lekkie rzeczy zmodyfikowaliśmy i mieliśmy gotową stronę gdzie z łatwością było o połączenie z baza danych.